  - Graylog Extended Log Format (GELF; https://go2docs.graylog.org/5-0/getting_in_log_data/gelf.html)
  - The Windows Event Log schema (http://mng.bz/d1AN), as a query format if you're using Windows Event Explorer
  - Common Event Format (CEF; https://docs.nxlog.co/userguide/integrate/cef-logging.html)
- In the time series databases realm or for the metrics signal type:
  - PromQL (http://mng.bz/qrGw), the Prometheus query language
  - The Prometheus exposition formats (http://mng.bz/rW7B) and OpenMetrics (https://openmetrics.io/)
  - InfluxData's Flux (https://www.influxdata.com/products/flux/), a standalone data scripting and query language
- For the traces signal type, there are no widely established query standards for traces yet, but Grafana Labs is working on this topic. Also, OpenTelemetry OTLP, as the default wire format, is widely adopted in this context.
- For signals of type profiles, `pprof` (https://github.com/google/pprof) is widely used; see also chapter 9, where we will dive deeper into this topic.

Note that for most of the aforementioned signal types (profiles excluded, expected to be available in 2023), the OpenTelemetry OTLP (http://mng.bz/V1KP) is already, or will soon be, the standard way to perform data exchange. In most cases, you want to choose a backend that supports an (open) industry standard, such as OpenTelemetry or OpenMetrics, over proprietary (only) ones, but see also the previous section on open standards, regarding the expected caveats in this context.

With this selection guide, we've reached the end of the chapter on backends. Next, we will discuss frontends (e.g., Grafana) and all-in-one solutions, such as Apache SkyWalking.

Summary

- Backends are the source of truth for your observability questions.
- The backends store signals in a durable and long-term manner, supporting various query languages, typically declarative.
- There are various offerings for logs, metrics, and traces backends. You can choose between what cloud providers offer, open source (and run-yourself) options, and managed offerings from a wide range of providers.
- Some backends (e.g., OpenSearch in the OSS space and most commercial offerings from observability vendors) can handle more than one signal type; however, in practice, one can end up with different backends per signal type.
- Being able to move from one signal type in a backend to another signal type in another—potentially different—backend is crucial to answer arbitrary questions.
- Cardinality explosion is the challenge that a metric dimension can take on many different values (think user ID).